# Zürich Tables

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The Zürich Tables are the internal source ledgers used at the Eidgenössische Sternwarte Zürich to construct the official daily sunspot number during the post-war era. They record, per observer and per day, the raw groups (G) and spots (S), the adopted Wolf/Zürich number R = k(10G+S), and notes on substitutions, weather/seeing, and instruments. The Waldmeier "primary" tables are especially informative about which value was used and why.

### Time span

1945–1979 (scanned sets ~2,000+ tables). Locating/encoding 1980 is a stated goal to bridge to the Brussels era (1981–).

### Zürich Tables are:

- Only day-level documentary trail of how Zürich actually derived R (observer selection, k-factors, alternates).
- Essential evidence for diagnosing and quantifying the 1947 discontinuity.

- Anchor for observer chaining and scaling in 1945–1979; supports uncertainty models and hemispheric series.
- Provides links to contemporaneous drawings (e.g., USET) for positional/area cross-checks.

## Current Status (FARSuN)

### Data retrieval & recovery

- ETH scans secured; internal working copies curated.
- The late-2010s recovery of Waldmeier tables closes a critical raw-data gap.

### Digitization

- Core G, S, R encoded for 1945–1979.
- Rich metadata (observer, instrument, notes) encoded for 1945–1959; remaining years in progress.
- Layout differences handled with tailored templates.
- Initial linkage to USET drawings underway.

### Quality control

- Normalizing observer identities/aliases; separating person / instrument / observatory in the DB.
- Cross-checking rubric/page vs. scans; anomaly flags (e.g., 1947 low coverage).
- Comparative audits of daily G, S, R; database views support change tracking.

### Metadata model

- Dedicated tables for person, instrument, observatory; draft condition/quality fields (method, seeing, projection).
- Provenance preserved (source IDs, pages, scan references).

### Accessibility

- Hosted in the historical sunspot observations database;
- Publication target: VO/EPN-TAP with standardized metadata;
- Citizen-science Streamlit prototype for transcription/validation.

## Next Steps (Focused, Actionable)

- Finish metadata pass (1951–1979) using the finalized condition schema.
- Confirm/ingest 1980 Zürich material (or document gap + statistical bridge).
- Publish an Observer Authority File (IDs, aliases, instruments, active periods).
- Automate provenance round-trip (cell-level image snippets, page hashes).
- Re-estimate k-factors for key observers with robust/Bayesian methods.
- Deep-dive 1947 using day-level substitutions and coverage diagnostics.
- Scale up drawing links (USET Zürich) to validate counts vs. areas/positions.
- Stage releases: (a) daily R + flags, (b) per-observer G,S,k + metadata, (c) derived calibration products, all via EPN-TAP/DOI.
- Citizen-science v2: pairwise validation, observer disambiguation, instrument tagging.